

TSFAdv: Frequency-Guided Black-Box Adversarial Attacks on Time Series Forecasting

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Abstract

While deep neural network-based long-term time series forecasting (LTSF) has become indispensable for critical infrastructures such as smart grids and IoT platforms, the deployment of these models as black-box APIs introduces severe security vulnerabilities that remain largely underexplored. In this paper, we propose TSFAdv, a query-efficient adversarial framework for LTSF models. The framework systematically analyzes model sensitivity to spectral perturbations in both magnitude and phase of the frequency domain. By embedding frequency-domain priors into Natural Evolution Strategies, we achieve sensitivity-guided gradient estimation that improves perturbation efficacy without violating practical query constraints. To overcome ambiguities inherent to point-wise regression metrics, we adopt a trajectory-level evaluation protocol based on Dynamic Time Warping (DTW) and Slope Misalignment Error (SME), enabling the capture of complex geometric and directional deviations. Extensive experiments across seven state-of-the-art architectures demonstrate that TSFAdv achieves substantial performance gains, with median DTW improvements of 38.78% and median SME improvements of 26.47% under 200-query budget. These findings reveal that existing defense mechanisms are ineffective against frequency-domain manipulation, underscoring an urgent necessity for robust LTSF models.¹

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¹Code: <https://github.com/HanQizhuo/TSFAdv>

1. Introduction

Deep neural network-based long-term time series forecasting (LTSF) has demonstrated significant utility in supporting modern critical infrastructure, assisting in essential tasks ranging from smart grid load balancing to climate monitoring (Wu et al., 2023; Wang et al., 2024a; Zeng et al., 2023; Liu et al., 2024; Nie et al., 2023; Zhou et al., 2022; Wu et al., 2021). The advancement of pre-trained foundation models has further facilitated their adoption in diverse production environments (Ansari et al., 2024; Das et al., 2024; Wang et al., 2025; Rasul et al., 2023), providing the accurate forecasts that aid in complex decision-making processes. However, the opacity of these black-box architectures presents potential security challenges, as the adversarial robustness of forecasting trajectories remains insufficiently explored.

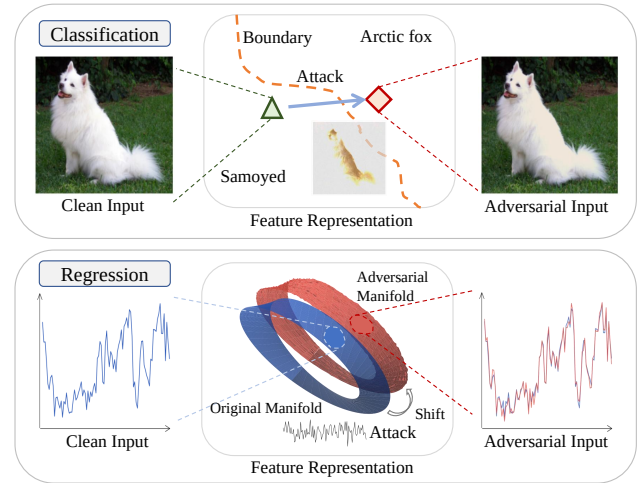


Figure 1. Adversarial Mechanisms: Classification vs. Regression. While classification (top) targets discrete boundary crossings, regression (bottom) requires manipulating continuous manifolds. This highlights the challenge in LTSF: overcoming the smoothness of the regression hyperplane to force significant trajectory shifts.

Research on adversarial robustness in black-box forecasting remains relatively unexplored, as existing paradigms are not well suited to deterministic LTSF. Transfer-based strategies necessitate prohibitive data access for surrogate training (Akhtar & Mian, 2018), while probabilistic methods (Liu et al., 2023) exhibit a structural incompatibility

with deterministic point-forecasting architectures. To the best of our knowledge, a systematic study of query-driven adversarial attacks against black-box LTSF models has not yet been reported. This gap is nontrivial, as small deviations may accumulate over long temporal horizons and lead to severe degradation in downstream systems.

While adversarial attacks are mature in classification (Cai et al., 2022; Oh et al., 2023; Ding et al., 2023; Karim et al., 2020; Fawaz et al., 2019; Yang et al., 2022), extending them to black-box LTSF encounters fundamental impediments: (1) **Temporal Redundancy**: LTSF models leverage long-range dependencies to suppress local perturbations, naturally mitigating attack efficacy. (2) **Manifold Continuity**: Unlike classification, which targets discrete decision boundaries for label flipping, LTSF operates on continuous regression manifolds (Figure 1). Successful attacks must overcome this inherent smoothness to induce significant trajectory deviations. (3) **Evaluation Ambiguity**: Quantifying attack success is non-trivial in regression, as distinguishing valid adversarial degradation from intrinsic stochastic forecasting errors or incidental accuracy fluctuations remains a persistent challenge.

To address these hurdles, we propose TSFAdv, a frequency-guided black-box framework exploiting the critical reliance of LTSF models on spectral characteristics (Zhou et al., 2022; Nie et al., 2023; Liu et al., 2024). Our preliminary study indicates that spectral perturbations tend to be more effective than temporal perturbations, suggesting underlying vulnerabilities. Built upon Natural Evolution Strategies (NES) (Ilyas et al., 2018), TSFAdv integrates two novel components—*offline amplitude-phase sensitivity profiling* and *sensitivity-weighted gradient estimation*—to bypass temporal redundancy and regression constraints. Furthermore, we introduce trajectory-level metrics to resolve evaluation ambiguities and evaluate attack effectiveness.

Our contributions are summarized as follows:

- This paper, for the first time, empirically demonstrates that frequency-domain perturbations can exploit spectral vulnerabilities that persist across forecasting horizons, thereby effectively mitigating the attenuation caused by temporal redundancy.
- We propose TSFAdv, a frequency-guided black-box adversarial attack method with offline profiling and sensitivity-weighted gradient estimation, enabling effective trajectory manipulation under strict query constraints and addressing regression complexity in LTSF attacks.
- To overcome ambiguities inherent to point-wise regression metrics, we introduce trajectory-level metrics for evaluation. Extensive experiments across seven state-of-the-art architectures demonstrate that TSFAdv achieves substantial performance gains, with average DTW improvements of 21.91-85.00% and SME improvements

of 15.04-61.97% under a restrictive 200-query budget, revealing critical spectral vulnerabilities in LTSF models.

Conflict of Interest Disclosure. The authors declare no financial or other conflicts of interest that could be perceived to influence this work.

2. Preliminaries

2.1. Threat Model

We consider a black-box adversarial setting where the forecasting model f is deployed as a service within networked devices. The attacker has no access to the model’s parameters, training data, or architecture. The attacker can only interact with the model through queries, i.e., by submitting time series $\mathbf{X}$ and receiving predictions $\hat{\mathbf{Y}} = f(\mathbf{X})$.

The attacker is allowed to perturb the input with an additive perturbation δ , yielding $\mathbf{X}' = \mathbf{X} + \delta$, where the perturbation is bounded by $\|\delta\|_p \leq \epsilon$. The goal of the attacker is to maximize the forecasting error, inducing significant deviations in the predicted trajectory while remaining within realistic perturbation constraints. Moreover, the number of queries available to the attacker is assumed to be limited, reflecting practical restrictions in real-world forecasting services. The attacker may additionally leverage a small set of domain-relevant public samples (e.g., publicly available energy records) for offline sensitivity profiling, requiring no access to the model’s proprietary training data. The practical viability of this assumption is empirically supported in Appendix C.6.

2.2. Preliminary study

Prior to introducing our method, we investigate the inefficacy of image classification attacks (Chen et al., 2017; Ilyas et al., 2018) within long-term time series forecasting (Liu et al., 2024; Wang et al., 2024a; Wu et al., 2023; Zeng et al., 2023). Unlike classification models that exploit sharp decision boundaries, forecasting models prioritize continuous temporal dependencies (Figure 1). As a result, small perturbations are absorbed by the model’s temporal smoothness and redundancy, failing to induce meaningful deviations in regression outputs. To investigate how perturbations across distinct representation domains influence forecasting behavior, we conduct empirical studies that systematically probe model responses to localized temporal and structured spectral perturbations. The results are presented in Figure 2.

Time-domain perturbation analysis. We inject Gaussian-shaped perturbations $\delta_{\text{gaussian}}(i) = A \cdot \exp\left(-\frac{(i-t)^2}{2\sigma^2}\right)$ with fixed amplitude $A = 0.01$ (normalized), sweeping center t and spread σ . Prediction deviations are quantified via RMSE, serving as a sufficient proxy for response magnitude in this sensitivity probing. As shown in the top row of

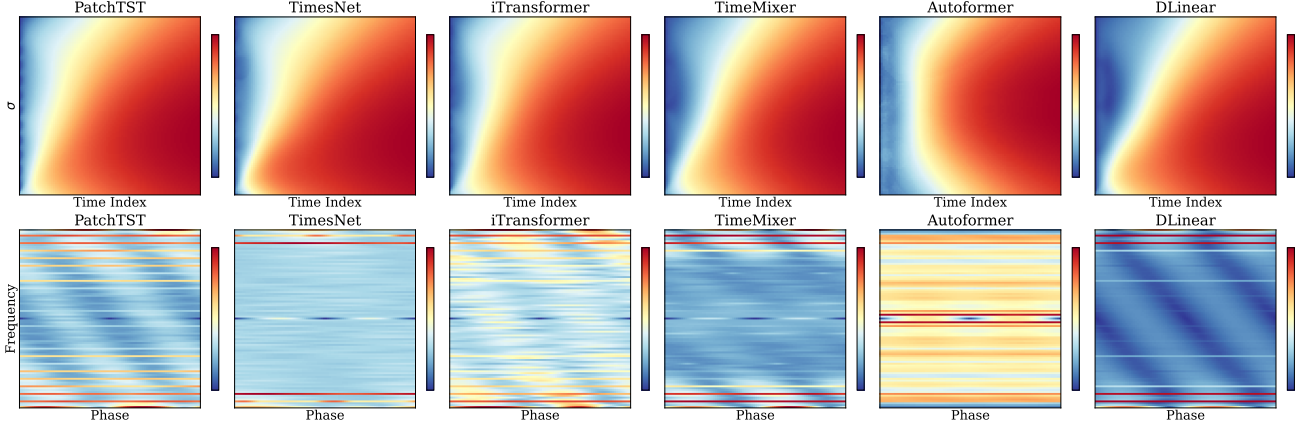


Figure 2. Comparative Sensitivity Landscapes. Error surfaces of six SOTA models under time-domain (Top) and frequency-domain (Bottom) perturbations. Red indicates high sensitivity. **Color scales are normalized individually per subplot** to elucidate local, architecture-specific vulnerability patterns; see Figure 7 for per-model unified-scale visualizations. Crucially, the sharp spectral peaks contrast with diffuse temporal responses, demonstrating that frequency-domain features exert a significantly stronger influence on forecasting outcomes.

Figure 2, responses display varying sensitivity to spread (σ) or location (t) but remain generally diffuse compared to the frequency domain, offering limited discriminative power for precise architecture-specific vulnerabilities.

Frequency-domain perturbation analysis. We apply sinusoidal perturbations $\delta_{\sin}(i) = A \sin(\omega i + \phi)$, varying frequency ω and phase ϕ while fixing A , and evaluate RMSE for each (ω, ϕ) pair. The bottom row of Figure 2 reveals sharp, architecture-specific peaks in the response surfaces, indicating that targeted spectral perturbations induce large forecasting errors with minimal time-domain magnitude, which exposes vulnerabilities invisible in the time domain.

3. Approach

We present TSFAdv, a frequency-domain adversarial attack framework for black-box LTSF models (Figure 3, Section A). Building on our empirical findings in Section 2.2, TSFAdv combines three key components: spectral perturbations targeting vulnerable amplitude-phase regions, offline sensitivity profiling to quantify architecture-specific dependencies, and sensitivity-weighted Natural Evolution Strategies for query-efficient optimization. Unlike time-domain methods susceptible to temporal redundancy attenuation, our approach exploits the inherent spectral sensitivity of modern LTSF architectures.

Notation. We denote the input time series as $\mathbf{X} \in \mathbb{R}^{T \times F}$, where T is the sequence length and F the number of channels. The forecasting model $f : \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^{H \times F}$ maps inputs to predictions over horizon H . $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts of a complex quantity, and $\theta \in \mathbb{R}^{2TF}$ denotes the vectorized frequency-domain optimization variable.

3.1. Sensitivity Profiling

Building on insights from Section 2.2, which revealed substantial impacts of perturbations at certain frequencies and phases, we systematically model these sensitivities through frequency-domain estimation. Motivated by techniques in computer vision (Yin et al., 2019) and time-series backdoor attacks (Huang et al., 2025), we characterize forecasting model responses to spectral perturbations, distinguishing between *amplitude sensitivity* and *phase sensitivity*, formally defined as follows:

Definition 3.1 (Frequency-Domain Amplitude and Phase Sensitivity). Let $x \in \mathbb{R}^T$ be a normalized univariate time series of length T , and let $f(\cdot)$ denote a black-box forecasting model. The discrete Fourier transform (DFT) yields

$$X[k] = \text{DFT}(x)[k] = A[k]e^{i\phi[k]}, \quad k = 0, 1, \dots, T-1,$$

where $A[k] \geq 0$ and $\phi[k] \in [-\pi, \pi)$ denote amplitude and phase, with $X[T-k] = \overline{X[k]}$ for real signals. We introduce targeted perturbations at frequency k :

$$\textbf{Amp: } \tilde{X}_{\text{amp}}^{(k)}[k] = (A[k] + \varepsilon) \cdot e^{i\phi[k]},$$

$$\textbf{Phase: } \tilde{X}_{\text{phase}}^{(k)}[k] = A[k] \cdot e^{i(\phi[k] + \varepsilon)}.$$

The amplitude and phase sensitivity at frequency k are defined as:

$$S_{\star}(k) = \varepsilon^{-1} \|f(\tilde{x}_{\star}^{(k)}) - f(x)\|_2, \quad \star \in \{\text{amp, phase}\},$$

where $\tilde{x}_{\star}^{(k)} = \text{DFT}^{-1}(\tilde{X}_{\star}^{(k)})$ is real-valued by conjugate symmetry.

Higher sensitivity values indicate greater model reliance on that spectral feature, signaling vulnerability to adversarial

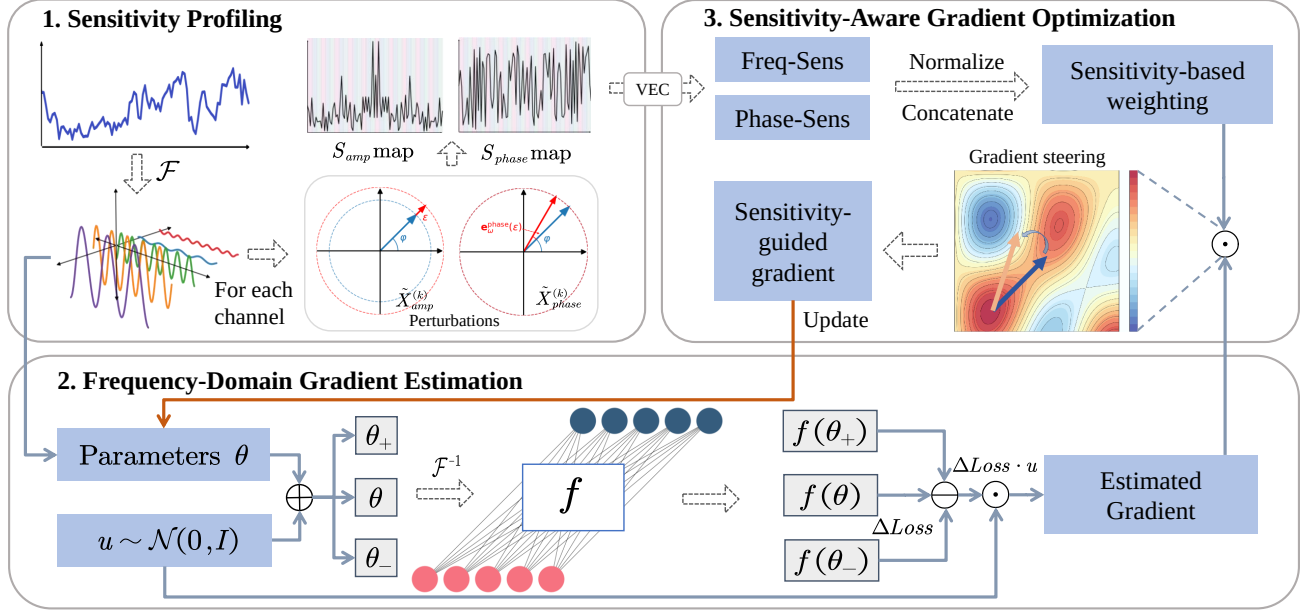


Figure 3. **TSFAdv attack pipeline:** (1) Fourier-based sensitivity estimation quantifies spectral vulnerabilities; (2) Natural Evolution Strategies estimate gradients via stochastic sampling; (3) Sensitivity-weighted optimization generates perturbations.

manipulation. Conjugate symmetry ensures the perturbed signals remain real-valued, maintaining signal validity while enabling precise measurement of spectral vulnerabilities. The resulting profiles $S_{amp}, S_{phase} \in \mathbb{R}^{\lfloor T/2 \rfloor \times F}$ provide architecture-specific spectral vulnerability maps, extended to full-spectrum maps $S_{amp}, S_{phase} \in \mathbb{R}^{T \times F}$ via $S(T - k) = S(k)$ for $k = 1, \dots, \lfloor T/2 \rfloor$.

To capture intrinsic, input-independent vulnerabilities, we estimate a *domain-level sensitivity profile* by averaging sensitivity maps over a small calibration set drawn from the same application domain. This one-time offline profiling is reused across all instances, rendering the cost negligible when amortized over persistent API usage. Consequently, per-instance complexity remains $\mathcal{O}(Q)$. For fairness, we explicitly include these profiling costs in our experimental budgets (Table 1) when comparing with baselines (Ilyas et al., 2018; Chen et al., 2017) (detailed cost analysis in Appendix C.5).

3.2. Frequency-Domain Gradient Estimation

While frequency-domain sensitivity provides valuable insights into a model’s susceptibility to distinct spectral components, translating such insights into actionable optimization guidance remains non-trivial. Conventional NES approaches overlook this structural prior when determining search directions, which results in suboptimal search efficiency. To overcome this limitation, we introduce a sensitivity-aware weighting scheme that leverages sensitivity as importance weights to modulate both sampling and gradient estimation, thereby facilitating more efficient and

targeted adversarial perturbation generation.

Frequency-domain parameterization. Given an input time series $X \in \mathbb{R}^{T \times F}$, where T denotes the length of the input sequence and F denotes the number of features (or channels), we apply the discrete Fourier transform (DFT) to obtain its spectral representation:

$$G = \text{DFT}(X) \in \mathbb{C}^{T \times F}, \quad G = \Re(G) + i \Im(G).$$

where $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts, respectively. The real and imaginary components are concatenated and vectorized to form the optimization variable:

$$\theta \triangleq \begin{bmatrix} \text{vec}(\Re(G)) \\ \text{vec}(\Im(G)) \end{bmatrix} \in \mathbb{R}^{2TF}$$

Gradient Estimation. To optimize adversarial perturbations in the frequency domain, we adopt the NES (Ilyas et al., 2018) framework and further integrate sensitivity as an importance prior. Specifically, we draw search directions $u_k \sim \mathcal{N}(0, I)$ and generate perturbed parameters $\theta_k^\pm = \theta \pm \sigma u_k$. After applying the inverse Fourier transform, the perturbed time series is reconstructed as $X_k^\pm = \Re(\text{DFT}^{-1}(\theta_k^\pm))$, and the forecasting loss is evaluated as the mean squared deviation from the clean prediction:

$$l_k^\pm = \frac{1}{|\Omega|} \|f(X_k^\pm) - y_0\|_2^2,$$

where $y_0 = f(X)$ represents the *clean prediction trajectory* of the unperturbed input, and $\Omega = \{1, \dots, H\}$ denotes the set of time indices in the forecasting horizon H .

3.3. Sensitivity-Aware Gradient Optimization

Sensitivity-Based Weighting. To incorporate structural priors of the frequency domain into the optimization process, we introduce a sensitivity-guided weighting scheme. Given the Frequency and phase sensitivities $S_{\text{amp}}, S_{\text{phase}} \in \mathbb{R}^{T \times F}$, we define a complex-valued sensitivity map: $S_c = S_{\text{amp}} + i S_{\text{phase}}$, where the modulus $|S_c|$ characterizes the overall vulnerability of each spectral component.

To align with the parameterization θ , we replicate the modulus across real and imaginary channels, and vectorize it as

$$w = \begin{bmatrix} \text{vec}(|S_c|) \\ \text{vec}(|S_c|) \end{bmatrix} \in \mathbb{R}^{2TF}.$$

Finally, we normalize the weights to remove scale bias:

$$\tilde{w} = \frac{w}{\bar{w}}, \quad \text{where} \quad \bar{w} = \frac{1}{2TF} \sum_{j=1}^{2TF} w_j.$$

This weighting mechanism biases the NES gradient estimation toward spectral regions where the model exhibits higher sensitivity, thereby enhancing both the efficiency and effectiveness of perturbation generation.

Gradient Estimation with Sensitivity Modulation. Unlike conventional NES which treats all frequencies equally, we introduce sensitivity-aware weighting to emphasize perturbations aligned with vulnerable spectral components. The weighted gradient estimator is thus given by

$$\hat{g} = \frac{1}{m} \sum_{k=1}^m \left(\frac{\ell_k^+ - \ell_k^-}{2\sigma} \right) (u_k \odot \tilde{w}),$$

where $\tilde{w}$ encodes the frequency-domain sensitivity. The estimated gradient is normalized as: $\hat{g} \leftarrow \frac{\hat{g}}{\|\hat{g}\|_2 + 10^{-8}}$, and the frequency-domain parameters are iteratively updated by: $\theta \leftarrow \theta + \eta \hat{g}$. The temporal perturbation is then reconstructed via the inverse transform:

$$\delta = \text{DFT}^{-1}(\theta).$$

This formulation explicitly leverages structural priors in the frequency domain to guide the search, thereby improving both the efficiency and the effectiveness of black-box adversarial optimization. The overall algorithm of TSFAdv is detailed in Appendix A.

4. Experiments

4.1. Experimental Setup

Datasets and Victim Models. We evaluate our method on five widely-used LTSF benchmarks: ETTh1, ETTh2, ETTm1, ETTm2 (Electricity Transformer Temperature

datasets with hourly and 15-minute granularities), and Weather (Wu et al., 2021). Following the standard protocol in Autoformer (Wu et al., 2021), we set both the input sequence length and prediction horizon to 96 for all experiments (detailed statistics in Appendix C.1). Our attack framework targets seven LTSF models spanning diverse design paradigms: linear models (DLinear (Zeng et al., 2023)), transformer-based models (Autoformer (Wu et al., 2021), PatchTST (Nie et al., 2023), iTransformer (Liu et al., 2024)), CNN models (TimesNet (Wu et al., 2023)), and hybrid models (TimeMixer (Wang et al., 2024a), TimeXer (Wang et al., 2024c)). (architectural specifications in Appendix C.1).

Baseline Attacker. Due to the lack of mature research in the related field, we adapt methods from computer vision, namely NES (Ilyas et al., 2018) and ZOO (Chen et al., 2017), to time series forecasting for baseline comparison, alongside random Gaussian noise generation (detailed formulations in Appendix C.3). All attackers are configured with identical hyperparameters for fair comparison: perturbation norm $\epsilon = 0.1$ under L_∞ constraint, noise level $\sigma = 0.025$ for sampling, and 10 optimization iterations. For our TSFAdv method, the sensitivity estimation step size is set to $\epsilon_{\text{sens}} = 0.01$ for both frequency and phase perturbations. A query limit of 200 is applied uniformly to all methods (cost analysis in Appendix C.5).

Evaluation Metrics. Conventional point-wise metrics (MSE, MAE) fail to capture temporal misalignment (Ratanamahatana & Keogh, 2004) and trajectory-level distortions (Ebert & McBride, 2000) essential for evaluating adversarial attacks on forecasting models. We adopt two complementary trajectory-level metrics (formal definitions in Appendix C.4).

- **Dynamic Time Warping (DTW):** Evaluates geometric deformation and phase misalignments via the optimal warping path π , defined as:

$$\min_{\pi} \sqrt{\sum_{(i,j) \in \pi} \|\hat{Y}_i - \hat{Y}'_j\|^2}.$$

- **Slope Misalignment Error (SME):** Quantifies directional trend reversals using the cosine similarity ρ of gradients $\Delta \hat{Y}$, formulated as:

$$\sqrt{(1 - \rho)/2} \in [0, 1].$$

Both metrics are computed between predictions $\hat{Y} = f(\mathbf{X})$ and perturbed predictions $\hat{Y}' = f(\mathbf{X}')$, where higher values indicate stronger attacks. DTW emphasizes geometric deformation while SME emphasizes directional misalignment, providing complementary assessments whose practical distinction is further illustrated in Appendix C.4.4.

TSFAdv: Frequency-Guided Black-Box Adversarial Attacks on Time Series Forecasting

Table 1. Main results: DTW and SME values (higher means better attack) across models, datasets, and attackers. *Imp* denotes the relative improvement between our method and the best-performing method other than TSFAdv (%).

Models	Attackers	ETTh1		ETTh2		ETTm1		ETTm2		Weather		Avg Imp	
		DTW ↑	SME ↑	DTW ↑	SME ↑	DTW ↑	SME ↑	DTW ↑	SME ↑	DTW ↑	SME ↑	DTW ↑	SME ↑
TimeMixer	Gaussian	15.7955	0.0649	11.1795	0.1267	13.8762	0.0961	13.0314	0.1416	14.6194	0.0596	50.45%	25.10%
	Zoo	15.3105	0.0634	10.7492	0.1208	13.0351	0.0888	12.0545	0.1234	4.4604	0.0229		
	Nes	18.7781	0.0652	18.3352	0.1277	20.0898	0.0972	20.2959	0.1435	48.6889	0.1072		
	TSFAdv	28.2254	0.0745	25.2553	0.1439	27.4888	0.1133	27.4417	0.1891	93.5792	0.1610		
	<i>Imp</i>	50.31%	14.26%	37.74%	12.69%	36.83%	16.56%	35.21%	31.78%	92.20%	50.19%		
iTransformer	Gaussian	8.2517	0.0263	8.9470	0.0657	14.6165	0.0554	13.0749	0.1116	75.9402	0.3328	28.15%	26.47%
	Zoo	11.1302	0.0161	16.0615	0.0510	22.8780	0.0566	28.1546	0.1232	84.7408	0.3374		
	Nes	23.4000	0.0321	27.2429	0.0877	38.7640	0.0958	41.9331	0.1934	93.5868	0.3853		
	TSFAdv	32.2841	0.0451	36.0618	0.0952	50.2864	0.141	47.6206	0.2263	118.981	0.459		
	<i>Imp</i>	37.97%	40.50%	32.37%	8.55%	29.72%	47.18%	13.56%	17.01%	27.13%	19.12%		
TimeXer	Gaussian	9.3140	0.0260	10.9404	0.0629	13.1133	0.0447	10.6341	0.0762	44.0844	0.1811	21.91%	24.29%
	Zoo	12.9508	0.0132	17.9083	0.0428	19.3451	0.0382	15.4661	0.0574	49.4699	0.1758		
	Nes	26.4371	0.0277	31.9019	0.0802	33.9879	0.0674	25.3239	0.1009	91.0254	0.2823		
	TSFAdv	28.7072	0.0347	38.6839	0.0922	36.4981	0.0821	31.3898	0.1295	135.0501	0.3700		
	<i>Imp</i>	8.59%	25.27%	21.26%	14.96%	7.38%	21.81%	23.95%	28.34%	48.36%	31.06%		
TimesNet	Gaussian	7.9684	0.0212	11.2366	0.0825	10.2341	0.0638	12.1304	0.1668	54.951	0.2083	28.79%	41.27%
	Zoo	15.6061	0.0305	24.1741	0.1237	17.3721	0.0628	28.0574	0.1864	88.0232	0.2237		
	Nes	33.5476	0.063	39.5224	0.1949	32.2006	0.1086	50.7559	0.2570	203.5007	0.3918		
	TSFAdv	50.5481	0.1115	47.8849	0.244	50.1566	0.1656	58.3965	0.3517	206.1333	0.4500		
	<i>Imp</i>	50.67%	76.98%	21.16%	25.19%	55.76%	52.49%	15.05%	36.84%	1.29%	14.85%		
DLinear	Gaussian	4.8623	0.0105	5.5483	0.0373	7.6185	0.0367	7.0663	0.0519	8.1449	0.0343	85.00%	61.97%
	Zoo	2.4375	0.0063	3.044	0.0149	3.3114	0.0158	3.5855	0.021	1.6229	0.0114		
	Nes	9.2728	0.0124	12.3389	0.0334	13.9518	0.0354	15.8831	0.0462	27.565	0.0647		
	TSFAdv	25.7566	0.0445	20.8296	0.0509	14.5333	0.0330	14.9799	0.0556	77.161	0.0767		
	<i>Imp</i>	177.76%	258.87%	68.81%	36.46%	4.16%	-10.08%	-5.69%	7.12%	179.92%	17.47%		
PatchTST	Gaussian	9.4536	0.0285	12.6720	0.0955	14.7470	0.0594	11.9013	0.0968	23.5282	0.1278	57.95%	33.10%
	Zoo	5.9481	0.0169	9.7896	0.0754	10.1601	0.0390	7.9689	0.0650	9.0083	0.0766		
	Nes	15.0244	0.0351	19.6840	0.1246	24.3183	0.0758	15.3953	0.1064	39.8582	0.1801		
	TSFAdv	22.1060	0.0424	30.4428	0.1557	31.993	0.1005	21.9197	0.1450	85.3111	0.2718		
	<i>Imp</i>	47.13%	20.80%	54.65%	24.96%	31.56%	32.58%	42.38%	36.27%	114.03%	50.92%		
Autoformer	Gaussian	9.2052	0.0491	9.8045	0.1142	27.7764	0.2319	10.5156	0.2519	32.228	0.3238	38.78%	15.04%
	Zoo	14.5533	0.0758	13.2929	0.1588	45.7637	0.2924	13.8521	0.3200	33.5400	0.3362		
	Nes	19.9432	0.1049	14.4767	0.1816	59.3821	0.3429	15.5200	0.3569	49.1762	0.4134		
	TSFAdv	33.1144	0.1214	22.1711	0.2325	65.4044	0.3941	20.903	0.3879	63.869	0.4458		
	<i>Imp</i>	66.04%	15.93%	53.15%	28.03%	10.14%	14.93%	34.68%	8.69%	29.88%	7.84%		

4.2. Attack Performance Comparison

Table 1 demonstrates that TSFAdv consistently outperforms baseline methods across diverse model-dataset combinations. Our frequency-aware guidance is particularly effective on architectures with inherent spectral vulnerabilities, such as DLinear and TimeMixer, achieving up to **179.92% relative DTW improvement** over NES on the Weather dataset. These gains highlight the method’s ability to exploit frequency-domain weaknesses often missed by conventional spatial attacks.

Analyzing absolute attack magnitudes reveals distinct robustness profiles. Certain architectures, like TimesNet, exhibit high absolute DTW/SME values despite more modest relative improvements (e.g., 28.79%), suggesting intrinsic susceptibility to perturbations regardless of the attack

method. Conversely, scenarios where baselines fail but TSFAdv succeeds (e.g., PatchTST on ETTh1) produce substantial relative gains, confirming our approach’s efficacy in regimes where black-box attacks have historically struggled.

Aggregated results (rightmost columns) confirm that our framework delivers robust performance across architectures. Minor negative deviations on DLinear (ETTm1/ETTm2) are negligible and attributable to the limited spectral diversity of those datasets interacting with DLinear’s linear temporal dependencies. Overall, TSFAdv demonstrates superior stability and effectiveness under stringent query constraints.

Figure 11 (Appendix D) presents a representative attack instance on the ETTh dataset. Within the forecast horizon, the clean forecast closely tracks the ground truth, whereas the adversarial forecast exhibits systematic directional devi-

ations across multiple cycles. In electricity load forecasting, such trend reversals could cause operators to anticipate rising demand precisely when load is declining, illustrating that the trajectory distortions quantified by DTW and SME correspond to practically meaningful degradation in forecasting utility.

4.3. Hyperparameter Analysis

Perturbation Budget. To evaluate scalability, we vary ϵ over 20 uniformly sampled points in $[0.01, 0.2]$ under a fixed 200-query constraint. As shown in Figure 4 (averaged across four ETT datasets), TSFAdv consistently outperforms baselines with widening gaps, demonstrating that sensitivity guidance effectively targets scalable spectral vulnerabilities. Crucially, attack effectiveness exhibits continuous growth through $\epsilon = 0.2$ without the plateau observed in baselines (e.g., ZOO beyond $\epsilon \approx 0.12$), indicating substantial remaining capacity in the frequency-domain attack surface. Quantitatively, doubling ϵ from 0.1 to 0.2 yields a 150% DTW improvement for TSFAdv (vs. 100% for NES) on TimeMixer, verifying superior gradient estimation efficiency. Corresponding SME results (Figure 10, Appendix C.7) mirror these scaling trends, confirming the query efficiency advantage holds under both metrics.

Query Budget. We examine the impact of query count on attack effectiveness by varying the budget from 20 to 1000 queries with fixed $\epsilon = 0.1$. As shown in Figure 5, TSFAdv achieves efficient convergence (70–80% of maximum effectiveness within 200 queries) across four architectures, substantially outpacing baseline methods. This query efficiency stems from sensitivity-guided sampling that concentrates gradient estimation on vulnerable spectral regions, reducing the number of oracle calls required for effective perturbation generation. The results validate that our approach maintains practical viability even under stringent API rate limits commonly encountered in real-world black-box scenarios. Corresponding SME results (Figure 9, Appendix C.7) exhibit consistent convergence patterns across all four architectures, confirming that the query efficiency advantage holds under both evaluation metrics.

4.4. Frequency-Domain Sensitivity Analysis of Generated Perturbations

To empirically validate the spectral characteristics of generated perturbations, we conduct Discrete Fourier Transform (DFT) decomposition on adversarial perturbations produced by competing methods using the ETTh1 dataset with the iTransformer architecture, extracting real and imaginary components for quantitative comparison against derived frequency-domain sensitivity profiles. As depicted in Figure 6, baseline approaches exhibit negligible correlation with sensitivity patterns, whereas TSFAdv demonstrates

pronounced alignment. This empirical concordance substantiates the efficacy of sensitivity-guided optimization in exploiting architecture-specific spectral weaknesses, as further validated by ablation studies (Section 4.5).

4.5. Ablation Study

Table 2. Ablation study on sensitivity guidance. DTW and SME values are reported as averages across five datasets (higher indicates stronger attack).

Model	V1 (w/o Sens.)		V2 (Random)		V3 (Full)	
	DTW	SME	DTW	SME	DTW	SME
TimeMixer	22.880	0.062	11.701	0.060	38.422	0.099
iTransformer	48.292	0.166	27.331	0.127	57.047	0.193
TimeXer	39.680	0.114	16.594	0.079	54.066	0.142
DLinear	15.738	0.038	8.214	0.039	30.652	0.052
TimesNet	75.373	0.209	27.550	0.129	85.316	0.255
PatchTST	21.802	0.098	15.963	0.082	38.355	0.143
Autoformer	44.549	0.261	20.445	0.164	50.179	0.306
Average	32.157	0.123	16.708	0.092	43.940	0.157

To validate the contribution of each component in TSFAdv, we conduct ablation experiments comparing three variants under identical conditions (200-query budget, $\epsilon = 0.1$):

- **V1 (w/o Sens.):** Standard NES in the frequency domain with uniform weights $\tilde{w} = \mathbf{1} \in \mathbb{R}^{2TF}$, isolating the effect of frequency-domain parameterization alone.
- **V2 (Random):** NES in the frequency domain with random weights $w_{\text{rand}} \sim \mathcal{N}(0, 1)$, normalized as $\tilde{w}_{\text{rand}} = |w_{\text{rand}}| / \text{mean}(|w_{\text{rand}}|)$, serving as an unstructured weighting control.
- **V3 (Full):** The complete TSFAdv with offline-profiled amplitude and phase sensitivity weights (Section 3.1).

Table 2 reports results averaged across five datasets. We analyze the contribution of each component as follows.

Frequency-domain parameterization. Comparing vanilla NES from Table 1 (avg. DTW 36.32, SME 0.143) with V1 (avg. DTW 32.157, SME 0.123) reveals that frequency-domain parameterization alone yields a modest reduction of 11.5% in DTW. This indicates that operating in the frequency domain does not directly enhance attack strength by itself, but provides the structural foundation necessary for sensitivity-guided optimization.

Sensitivity-guided weighting. V3 outperforms V1 by +37.0% in DTW and +27.6% in SME, directly quantifying the contribution of structured sensitivity weights over uniform ones. The gap is especially pronounced on spectral-aware architectures such as Autoformer and iTransformer,

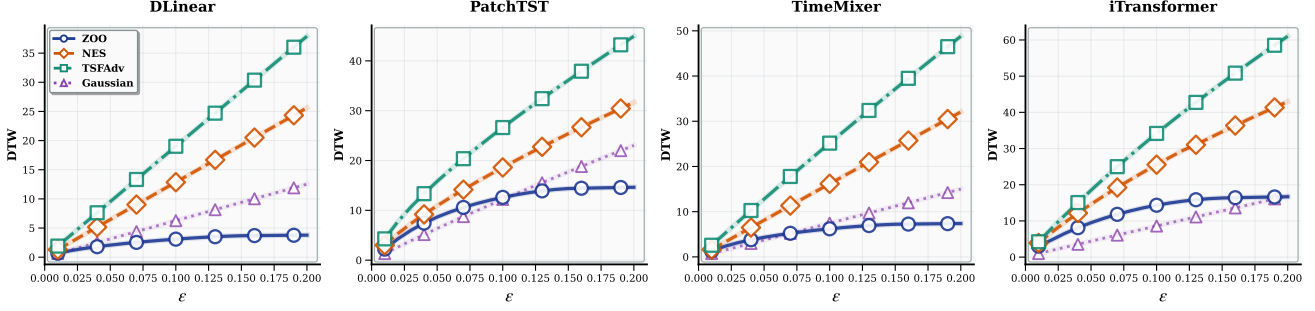


Figure 4. **Attack Effectiveness and Saturation Analysis.** Attack across varying perturbation budgets ($\epsilon \in [0.01, 0.2]$) under 200 queries. TSFAdv consistently outperforms baselines with steeper scaling trends and no saturation.

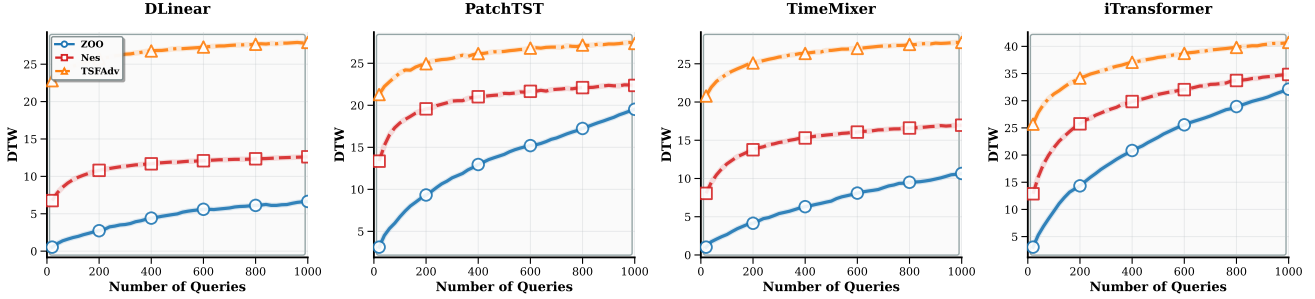


Figure 5. **Attack Convergence under Strict Query Constraints.** Attack performance versus query budget (20–1000 queries, $\epsilon = 0.1$). Metrics averaged across ETTh datasets.

Table 3. **Defense Effectiveness Against TSFAdv Attacks.** The attack performance under three representative defense mechanisms, compared to the **no-defense baseline**.

Models	Metrics	No Defense	Smoothing	Pruning	Adv-Train
DLinear	DTW	30.652	33.711	30.800	30.581
	SME	0.052	0.124	0.031	0.053
iTransformer	DTW	57.047	60.667	58.118	52.416
	SME	0.193	0.266	0.136	0.166
PatchTST	DTW	38.355	45.615	40.473	48.819
	SME	0.143	0.213	0.108	0.083
TimeMixer	DTW	40.398	40.584	38.422	38.529
	SME	0.136	0.142	0.099	0.103
TimeXer	DTW	54.066	49.106	45.697	49.507
	SME	0.142	0.204	0.089	0.134

where sensitivity guidance enables precise targeting of vulnerable amplitude-phase components. Furthermore, V2 performs significantly worse than even the uniform baseline V1, confirming that the structured sensitivity prior—rather than weight diversity—drives the performance gains. Together, these results demonstrate that the two components are complementary: frequency-domain parameterization establishes the optimization space, while sensitivity-guided weighting exploits the architecture-specific spectral structure within it.

4.6. Defense Mechanism

We evaluate three defense strategies against TSFAdv: **Randomized Smoothing**, **Magnitude-based Pruning**, and **Ad-**

versarial Training (detailed implementation configurations are provided in Appendix E). Table 3 reveals that these mechanisms exhibit limited effectiveness against frequency-domain attacks. Notably, randomized smoothing increases errors (e.g., +38.2% SME) by amplifying spectral perturbations, while pruning and adversarial training yield inconsistent or erratic mitigation. These findings underscore a critical gap: defenses designed for time-domain perturbations fail to effectively transfer to the frequency-domain vulnerabilities of forecasting models.

5. Related Work

5.1. DNN-Based LTSF

Deep neural networks (DNNs) have established dominance in Long-Term Time Series Forecasting (LTSF), superseding statistical methods via superior modeling of complex dependencies in critical domains (Wu et al., 2023; Wang et al., 2024a; Zeng et al., 2023; Liu et al., 2024; Nie et al., 2023; Zhou et al., 2022; Wu et al., 2021). The current state-of-the-art landscape features diverse architectural paradigms, ranging from linear projections (DLinear (Zeng et al., 2023)) and attention mechanisms (iTransformer (Liu et al., 2024), TimesNet (Wu et al., 2023)) to hybrid spectral-temporal frameworks (FEDFormer (Zhou et al., 2022), TimeMixer (Wang et al., 2024a)), all demonstrating enhanced scalability for production deployment.

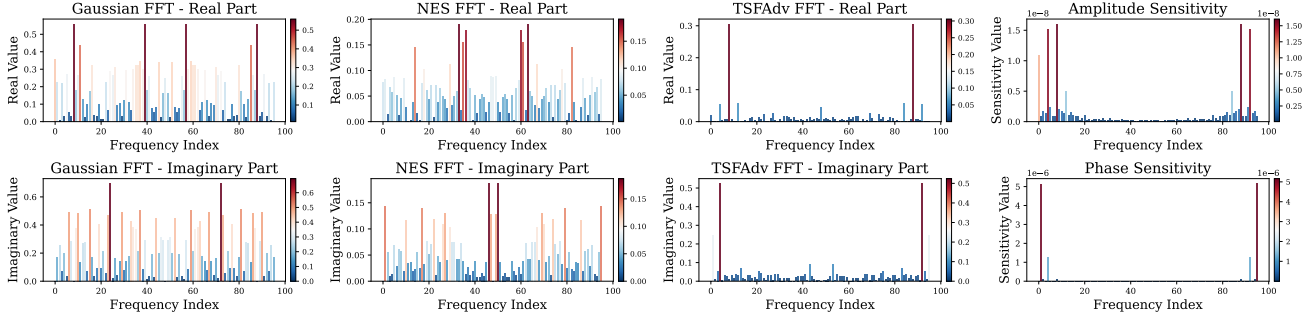


Figure 6. **Spectral Alignment Verification.** Frequency-domain sensitivity analysis comparing perturbation methods. Our proposed TSFAdv demonstrates superior alignment with frequency-domain sensitivity patterns compared to baseline approaches.

5.2. Black-Box Adversarial Attack

Black-box attacks, pioneered in computer vision (Papernot et al., 2017; Chen et al., 2017; Ilyas et al., 2018), aim to craft input perturbations without access to model internals, relying solely on input-output interactions. As research progressed, black-box attack methodologies (Cai et al., 2022; Oh et al., 2023; Qin et al., 2022; Wei et al., 2023; Lee et al., 2025; Shi et al., 2022; Kim et al., 2023a,b; Aich et al., 2022; Tarchoun et al., 2023) in computer vision have evolved significantly, both in terms of query efficiency and perturbation realism. This line of research has recently been extended to time series, primarily in the context of time series classification (TSC) (Ding et al., 2023; Karim et al., 2020; Fawaz et al., 2019; Yang et al., 2022).

In contrast, LTSF presents unique challenges regarding adversarial robustness. While modern architectures like Autoformer (Wu et al., 2021) and iTransformer (Liu et al., 2024) are widely deployed as black-box APIs, their security in such settings remains underexplored. Existing research is largely inapplicable, as it relies on white-box assumptions (Krishan et al., 2024), or focuses on probabilistic frameworks (Liu et al., 2023). These approaches are ill-suited for mainstream LTSF services: transfer-based methods demand prohibitive data and query budgets for surrogate training, while probabilistic attacks are incompatible with deterministic point estimation. Consequently, current strategies fail to address the constraints of query-limited, black-box forecasting systems.

5.3. Frequency-Domain Analysis

Frequency-domain analysis (Zeng et al., 2021; Wang et al., 2018; Chen et al., 2024; Zhou et al., 2022; Sun & Boning, 2022; Woo et al., 2022) has proven pivotal for understanding model robustness. In vision, Yin et al. (Yin et al., 2019) exploited high-frequency sensitivity for attacks, a concept recently extended to time series classification via spectral backdoors (Huang et al., 2025). However, the latter relies on training-time access, differing fundamentally

from our black-box inference setting. Although modern LTSF architectures like TimesNet (Wu et al., 2023) and TimeMixer (Wang et al., 2024a) leverage spectral representations to model dependencies, their resulting vulnerability to query-efficient spectral attacks remains unexplored. We address this by proposing a framework that jointly exploits amplitude and phase structures to attack deterministic LTSF models under strict query constraints. Crucially, we incorporate a sensitivity profiling mechanism to guide gradient estimation, thereby revealing architecture-specific vulnerabilities invisible to time-domain baselines.

6. Conclusion

This paper introduces TSFAdv, a frequency-guided black-box adversarial framework designed to expose spectral vulnerabilities in long-term time series forecasting (LTSF). By integrating offline amplitude-phase sensitivity profiling with Natural Evolution Strategies, TSFAdv enables query-efficient perturbation generation under strict API constraints. Extensive evaluations across seven architectures and five datasets demonstrate substantial performance gains, achieving 21.91–85.00% DTW and 15.04–61.97% SME improvements over baselines within a 200-query budget, validated through trajectory-level metrics that capture structural deformations. Crucially, our analysis reveals that existing defenses are largely ineffective against frequency-domain attacks, sometimes inadvertently amplifying vulnerabilities, highlighting an urgent need for robust, frequency-aware defense mechanisms in critical forecasting infrastructures.

Several directions remain open for future investigation. Extending TSFAdv to longer prediction horizons and more diverse deployment scenarios would broaden its applicability. Developing frequency-aware defense mechanisms tailored specifically to LTSF models represents a pressing next step motivated by our findings. Additionally, exploring the calibration-set sensitivity across more diverse domains and the transferability of sensitivity profiles across model families could further reduce profiling overhead in large-scale deployment settings.

Acknowledgements

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Impact Statement

This paper reveals critical spectral vulnerabilities in long-term time series forecasting models. While our frequency-domain attack framework demonstrates how such systems could be compromised, our primary goal is to provide a rigorous tool for robustness evaluation. By highlighting these risks, we aim to catalyze the development of frequency-aware defense mechanisms for sensitive applications like smart grids and IoT platforms, ultimately enhancing the security and reliability of web-scale predictive infrastructure.

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A. Comprehensive TSFAdv Algorithm

We formally present the complete execution procedure of TSFAdv in Algorithm 1. To reconcile the requirement for high attack success rates with stringent query constraints, the framework is architected into two decoupled phases: (1) **Offline Sensitivity Profiling**, which extracts model-intrinsic spectral vulnerabilities using a representative calibration set; and (2) **Online Sensitivity-Guided Perturbation**, which leverages these pre-computed priors to guide the gradient estimation for specific target instances. This separation ensures that the computational overhead of sensitivity analysis is amortized across all subsequent attacks, thereby preserving the query efficiency of the online phase.

Algorithm 1 TSFAdv: Frequency-Aware Black-Box Attack

Require: Target Model f , Target Input $\mathbf{X} \in \mathbb{R}^{T \times F}$, Calibration Set $\mathcal{D}_{\text{cal}} = \{\mathbf{X}_1, \dots, \mathbf{X}_N\}$, Budget ϵ , Queries Q , Noise σ , Samples m

Ensure: Adversarial input $\mathbf{X}^* = \mathbf{X} + \delta^*$

- 1: **Phase 1: Offline Sensitivity Profiling (Domain-Level Prior Extraction)**
- 2: {Computed once on $\mathcal{D}_{\text{cal}}$; addressing conjugate symmetry for real-valued signals.}
- 3: Initialize accumulators $\bar{S}_{\text{amp}}, \bar{S}_{\text{phase}} \leftarrow \mathbf{0}$
- 4: **for** each calibration sample $\mathbf{X}_c \in \mathcal{D}_{\text{cal}}$ **do**
- 5: $\mathbf{G} \leftarrow \text{DFT}(\mathbf{X}_c)$
- 6: **for** $k = 0$ to $\lfloor T/2 \rfloor$ **do**
- 7: {Apply perturbation to index k }
- 8: $\tilde{\mathbf{G}}_{\text{amp}}^{(k)}[k] \leftarrow (|\mathbf{G}[k]| + \epsilon_{\text{sens}})e^{i\angle \mathbf{G}[k]}$
- 9: $\tilde{\mathbf{G}}_{\text{phase}}^{(k)}[k] \leftarrow |\mathbf{G}[k]|e^{i(\angle \mathbf{G}[k] + \epsilon_{\text{sens}})}$
- 10: {Enforce conjugate symmetry at $T - k$ }
- 11: **if** $k > 0$ **and** $k < T/2$ **then**
- 12: $\tilde{\mathbf{G}}_{\text{amp}}^{(k)}[T - k] \leftarrow \tilde{\mathbf{G}}_{\text{amp}}^{(k)}[k]; \quad \tilde{\mathbf{G}}_{\text{phase}}^{(k)}[T - k] \leftarrow \overline{\tilde{\mathbf{G}}_{\text{phase}}^{(k)}[k]}$
- 13: **end if**
- 14: $\bar{S}_{\text{amp}}(k) \leftarrow \bar{S}_{\text{amp}}(k) + \frac{1}{|\mathcal{D}_{\text{cal}}|} \|f(\text{DFT}^{-1}(\tilde{\mathbf{G}}_{\text{amp}}^{(k)})) - f(\mathbf{X}_c)\|_2 / \epsilon_{\text{sens}}$
- 15: $\bar{S}_{\text{phase}}(k) \leftarrow \bar{S}_{\text{phase}}(k) + \frac{1}{|\mathcal{D}_{\text{cal}}|} \|f(\text{DFT}^{-1}(\tilde{\mathbf{G}}_{\text{phase}}^{(k)})) - f(\mathbf{X}_c)\|_2 / \epsilon_{\text{sens}}$
- 16: **end for**
- 17: **end for**
- 18: $S_{\text{amp}} \leftarrow \bar{S}_{\text{amp}}, \quad S_{\text{phase}} \leftarrow \bar{S}_{\text{phase}}$
- 19: **Phase 2: Online Sensitivity-Guided Perturbation (Targeting $\mathbf{X}$)**
- 20: $\mathbf{w} \leftarrow \text{vec}([S_{\text{amp}}; S_{\text{phase}}]); \tilde{\mathbf{w}} \leftarrow \mathbf{w} / \text{mean}(\mathbf{w})$
- 21: Initialize $\delta \leftarrow \mathbf{0}, \hat{\mathbf{Y}}_0 \leftarrow f(\mathbf{X}), \theta_\delta \leftarrow \text{vec}(\text{DFT}(\delta))$
- 22: **for** $t = 1$ to $\lfloor Q/(2m) \rfloor$ **do**
- 23: **for** $i = 1$ to m **do**
- 24: $\mathbf{u}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}); \theta_i^\pm \leftarrow \theta_\delta \pm \sigma \mathbf{u}_i$
- 25: $\ell_i^\pm \leftarrow \|\hat{\mathbf{Y}}_0 - f(\mathbf{X} + \text{Re}(\text{DFT}^{-1}(\text{reshape}(\theta_i^\pm))))\|_2^2$
- 26: **end for**
- 27: {Step 1: Aggregate raw gradients}
- 28: $\mathbf{g}_{\text{raw}} \leftarrow \frac{1}{m\sigma} \sum_{i=1}^m (\ell_i^+ - \ell_i^-)(\mathbf{u}_i \odot \tilde{\mathbf{w}})$
- 29: {Step 2: Normalize (Outside summation as per Sec 3.3)}
- 30: $\hat{\mathbf{g}} \leftarrow \mathbf{g}_{\text{raw}} / (\|\mathbf{g}_{\text{raw}}\|_2 + 10^{-8})$
- 31: $\theta_\delta \leftarrow \theta_\delta + \eta \hat{\mathbf{g}}; \delta \leftarrow \text{clip}(\text{Re}(\text{DFT}^{-1}(\text{reshape}(\theta_\delta))), -\epsilon, \epsilon)$
- 32: **end for**
- 33: **return** $\mathbf{X}^* = \mathbf{X} + \delta$

As formally outlined in Algorithm 1, the execution pipeline is structured into two logically distinct phases.

In **Phase 1** (Lines 1–17), the sensitivity maps S_{amp} and S_{phase} are derived as expectations over the calibration set $\mathcal{D}_{\text{cal}}$. Crucially, the algorithm enforces **conjugate symmetry** (Lines 9–11) during spectral perturbation. This step guarantees that the inverse DFT yields mathematically valid real-valued time series, addressing the theoretical constraint $\tilde{\mathbf{G}}[T - k] = \overline{\tilde{\mathbf{G}}[k]}$ inherent to real-signal processing.

In **Phase 2** (Lines 18–31), these global priors are employed to guide the online optimization. The search direction is modulated via the Hadamard product $\mathbf{u}_i \odot \tilde{\mathbf{w}}$ (Line 26), biasing the Natural Evolution Strategy (NES) towards spectrally vulnerable components. Furthermore, the gradient estimation is explicitly **decoupled** into raw aggregation (Line 26) and subsequent normalization (Line 29). This separation ensures that the normalization is applied to the estimated expectation of the gradient rather than individual samples, strictly adhering to the standard NES formulation and ensuring numerical stability.

B. Experimental Configuration and Analysis of Preliminary Sensitivity Studies

In this section, we provide a comprehensive exposition of the experimental protocols, mathematical formulations, and methodological rationales governing the preliminary sensitivity analysis presented in Section 2.2. The primary objective of this empirical study is to systematically characterize the distinct vulnerability surfaces of Long-Term Time Series Forecasting (LTSF) architectures when subjected to localized temporal anomalies versus global spectral perturbations.

Perturbation Protocols. To ensure a rigorous and fair comparison between the two perturbation modalities, we enforce a strictly consistent energy budget across all experiments. Let $\mathbf{X} \in \mathbb{R}^{T \times C}$ denote the input time series with sequence length T and channel dimension C .

For the **time-domain sensitivity analysis**, we simulate localized temporal irregularities by injecting a Gaussian-shaped perturbation. The perturbed input $\mathbf{X}'$ at time step t is formulated as:

$$\mathbf{X}'(t) = \mathbf{X}(t) + \delta_{\text{time}}(t; t_c, \sigma), \quad \text{where} \quad \delta_{\text{time}}(t; t_c, \sigma) = A \cdot \exp\left(-\frac{(t - t_c)^2}{2\sigma^2}\right).$$

Here, $t_c \in [0, T]$ determines the temporal center of the anomaly, and $\sigma \in [1, T/4]$ controls the spread (or duration) of the disturbance. We perform a high-resolution grid search over the parameter space (t_c, σ) to map the temporal sensitivity landscape.

Conversely, for the **frequency-domain sensitivity analysis**, we investigate the model’s response to specific spectral components using sinusoidal perturbations. The perturbation is defined as:

$$\mathbf{X}'(t) = \mathbf{X}(t) + \delta_{\text{freq}}(t; \omega, \phi), \quad \text{where} \quad \delta_{\text{freq}}(t; \omega, \phi) = A \cdot \sin(\omega t + \phi).$$

In this formulation, ω represents the angular frequency target, and $\phi \in [0, 2\pi)$ denotes the phase shift. We sweep ω across the discrete Fourier frequencies and sample ϕ uniformly. Crucially, in both scenarios, the amplitude A is fixed to a normalized constant (e.g., 0.01) to guarantee that the ℓ_∞ norm of the perturbation remains bounded and imperceptible, isolating the impact of perturbation *structure* rather than magnitude.

Rationale for Metric Selection (RMSE). While the core evaluation of our proposed TSFAdv framework employs trajectory-aware metrics (DTW and SME) to rigorously assess structural deformation and directional misalignment, this preliminary study deliberately utilizes the **Root Mean Square Error** as the response proxy. This methodological choice is grounded in two key considerations:

$$\text{RMSE}(\hat{\mathbf{Y}}, \hat{\mathbf{Y}}') = \sqrt{\frac{1}{H \cdot C} \sum_{i=1}^H \sum_{j=1}^C (\hat{\mathbf{Y}}_{i,j} - \hat{\mathbf{Y}}'_{i,j})^2}.$$

First, **Computational Efficiency for Landscape Profiling:** The preliminary analysis necessitates an exhaustive grid search over thousands of parameter combinations (e.g., varying ω and ϕ) to generate high-resolution sensitivity heatmaps. RMSE, with a computational complexity of $O(H)$, is significantly more efficient than Dynamic Time Warping (DTW), which scales quadratically $O(H^2)$. This efficiency allows for rapid and extensive probing of the vulnerability surface. Second, **Stability for Sensitivity Quantification:** The objective of this phase is to measure the raw *magnitude* of error amplification triggered by specific inputs, rather than to evaluate the morphological quality of the forecast. RMSE provides a deterministic and low-variance scalar signal that directly reflects instantaneous deviation. Unlike alignment-based metrics (DTW), which might “forgive” certain shifts, RMSE rigidly penalizes any deviation, making it a highly sensitive indicator for detecting “resonance” points (blind spots) in the model’s spectral processing.

Empirical Observations. The resulting sensitivity heatmaps (visualized in Figure 2) demonstrate a fundamental dichotomy in model robustness. The time-domain response surface is characterized by diffuse, low-gradient regions, suggesting that modern LTSF models—often employing attention mechanisms or pooling—possess inherent inertia against local temporal noise. In sharp contrast, the frequency-domain landscape exhibits sparse, high-magnitude peaks. These peaks correspond to specific “vulnerable frequencies” where the model’s internal spectral bias amplifies perturbations disproportionately. This observation validates our hypothesis that frequency-domain attacks can achieve higher effectiveness with lower query costs by targeting these specific spectral vulnerabilities.

Unified Color Scale Visualization. Figure 7 presents the same sensitivity landscapes as Figure 2 in the main text, but with unified color scales where the time-domain and frequency-domain rows share the same colorbar per model. Under this unified scaling, the time-domain row exhibits uniformly low absolute sensitivity across all six architectures, while the frequency-domain row shows substantially higher absolute responses with clearly localized peaks, directly confirming that frequency-domain perturbations induce significantly larger absolute model responses.

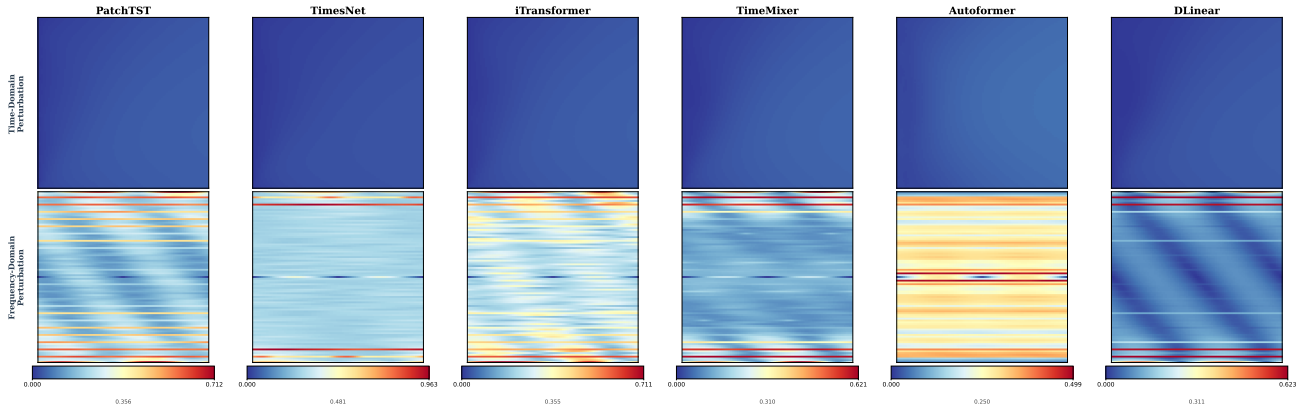


Figure 7. Sensitivity landscapes with **per-model unified color scales**. The time-domain (Top) and frequency-domain (Bottom) rows share the same colorbar for each architecture, directly revealing the substantially higher absolute sensitivity to frequency-domain perturbations.

C. Detailed Experimental Settings

C.1. Dataset and Model Specifications

C.1.1. DATASET DESCRIPTIONS

We conduct experiments on five benchmark datasets representing diverse real-world forecasting scenarios. The ETT (Electricity Transformer Temperature) datasets (Zhou et al., 2021) comprise multivariate time series from electricity transformers spanning July 2016 to July 2018. ETTh1 and ETTh2 provide hourly measurements with 17,420 time steps, while ETTm1 and ETTm2 offer 15-minute granularity with 69,680 time steps. All ETT variants contain 7 features including oil temperature and load measurements. The Weather dataset (Wu et al., 2021) contains 21 meteorological indicators recorded at 10-minute intervals throughout 2020, totaling 52,696 time steps. We adopt the standard 12:4:4 month train/validation/test split for ETT datasets and the official 70:10:20 split for Weather. All datasets undergo z-score normalization computed on training statistics. Following standard LTSF protocols (Wu et al., 2021), we configure input sequence length $T = 96$ and prediction horizon $H = 96$, performing direct multi-step univariate forecasting where each channel is predicted independently.

C.1.2. TARGET MODEL ARCHITECTURES

We evaluate seven LTSF models representing diverse design paradigms (Wang et al., 2024b). DLinear (Zeng et al., 2023) serves as the linear baseline by decomposing time series into trend-seasonal components and applying independent projections. Transformer-based models include Autoformer (Wu et al., 2021) with auto-correlation mechanisms for frequency-domain aggregation, PatchTST (Nie et al., 2023) with patch-wise channel-independent attention, and iTransformer (Liu et al., 2024) inverting attention across variates. TimesNet (Wu et al., 2023) represents CNN approaches by transforming 1D series into 2D period-temporal tensors. Hybrid architectures include TimeMixer (Wang et al., 2024a) employing multi-scale mixing

operations and TimeXer (Wang et al., 2024c) incorporating exogenous variables via cross-attention. Autoformer, TimesNet, and TimeMixer exhibit explicit frequency-aware designs. All models use official implementations with hyperparameters from (Wang et al., 2024b), early stopping (patience=10), and fixed seeds (seed=2024).

C.2. Attack Configuration and Query Budget Protocol

To evaluate the practical threat of adversarial attacks against deployed forecasting services, we define the query constraints based on the distinction between *setup costs* and *inference-time latency*.

Online Query Budget (Q_{online}). We strictly enforce a uniform budget of $Q_{\text{online}} = 200$ queries for the *online perturbation generation phase* across all methods (TSFAdv, NES, ZOO). This constraint simulates the strict rate limits and latency requirements of real-time web-scale systems, where an attacker has limited opportunities to interact with the API for a specific target instance before detection or timeout.

Fairness and Amortized Cost Analysis. While TSFAdv incurs a fixed initialization cost Q_{sens} for offline sensitivity profiling (approx. 33,600 queries; see Appendix C.5), we maintain that the comparison with zero-shot baselines (NES, ZOO) is methodologically sound based on the principle of amortized complexity and realistic threat modeling:

- **Amortized Complexity:** Sensitivity profiling is a model-specific, one-time operation decoupled from the instance-specific attack loop. When amortized across the extensive test sets typical of LTSF benchmarks (e.g., $\sim 2,800$ instances for ETT), the marginal query overhead is negligible (≈ 11.8 queries per instance). This represents a tractable setup cost that diminishes asymptotically with the scale of the attack campaign.
- **Threat Model Alignment:** In practical black-box scenarios, attackers frequently leverage offline reconnaissance to minimize online detection risks. TSFAdv formalizes this *prior-guided* paradigm, leveraging offline computation to maximize query efficiency during the critical, rate-limited online attack window, in contrast to the *blind* search employed by baselines.

Consequently, the evaluation metrics in Table 1 isolate and assess **online query efficiency**: the algorithm’s efficacy in identifying adversarial perturbations within a strictly constrained 200-query window, conditioned on the availability of pre-computed spectral priors.

C.3. Baseline Attacker

We selected NES (Ilyas et al., 2018) and ZOO (Chen et al., 2017) as baselines because no established black-box adversarial attack methods exist for long-term time series forecasting. These domain-agnostic methods are directly transferable to regression tasks, whereas alternatives like CMA-ES (Hansen et al., 2003) are fundamentally incompatible with query-limited scenarios: CMA-ES requires $O(T^2)$ queries per iteration, meaning a single iteration for $d = 672$ dimensions would demand $\sim 450,000$ queries, over 2,000 times our 200-query budget, making even one optimization step infeasible.

C.3.1. NATURAL EVOLUTION STRATEGIES

Natural Evolution Strategies (NES) (Ilyas et al., 2018) constitute a class of black-box optimization algorithms that approximate gradients by optimizing over a parameterized search distribution. The core idea is to reformulate the objective function $f(\cdot)$ as an expectation under a distribution $p_\theta(z)$ parameterized by θ :

$$J(\theta) = \mathbb{E}_{z \sim p_\theta(z)}[f(z)].$$

Using the score function estimator (also known as the log-likelihood ratio trick), the gradient of $J(\theta)$ with respect to θ is expressed as:

$$\nabla_\theta J(\theta) = \mathbb{E}_{z \sim p_\theta(z)}[f(z) \nabla_\theta \log p_\theta(z)].$$

In practice, when p_θ is instantiated as an isotropic Gaussian distribution $\mathcal{N}(\theta, \sigma^2 I)$, the gradient can be estimated via Monte Carlo sampling. Let N denote the number of sampled perturbations, and let $\epsilon_i \sim \mathcal{N}(0, I)$ for $i = 1, \dots, N$. The gradient estimator is then given by:

$$\nabla_\theta J(\theta) \approx \frac{1}{\sigma N} \sum_{i=1}^N f(\theta + \sigma \epsilon_i) \epsilon_i.$$

This formulation enables gradient estimation using only function evaluations, without requiring access to the model’s internal parameters or architecture. In the context of adversarial attacks, NES is particularly well-suited for black-box settings: an adversary perturbs an input $\mathbf{X}$ with additive noise δ , queries the forecasting model $f(\mathbf{X} + \delta)$, and uses the returned outputs to estimate the gradient direction. This query-based mechanism facilitates the construction of adversarial examples even when the target model is entirely opaque.

C.3.2. ZEROth ORDER OPTIMIZATION

Zeroth Order Optimization

(ZOO) (Chen et al., 2017) is a gradient-free black-box attack method that estimates gradients through finite differences using only function queries. The core idea is to approximate the partial derivative of the objective function $f(\cdot)$ with respect to each coordinate i of the input $\mathbf{x}$ by evaluating function values at perturbed points:

$$\frac{\partial f(\mathbf{x})}{\partial x_i} \approx \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x} - h\mathbf{e}_i)}{2h},$$

where h is a small perturbation magnitude and $\mathbf{e}_i$ is the unit vector along coordinate i . For an n -dimensional input, the full gradient is constructed by sequentially estimating all n partial derivatives:

$$\nabla f(\mathbf{x}) \approx \left[\frac{f(\mathbf{x} + h\mathbf{e}_1) - f(\mathbf{x} - h\mathbf{e}_1)}{2h}, \dots, \frac{f(\mathbf{x} + h\mathbf{e}_n) - f(\mathbf{x} - h\mathbf{e}_n)}{2h} \right]^\top.$$

This coordinate-wise gradient estimation requires $2n$ queries per iteration, making it computationally expensive for high-dimensional inputs. To improve efficiency, ZOO employs dimensionality reduction techniques and importance sampling to prioritize coordinates that contribute most to the attack objective. In the context of adversarial attacks on time series forecasting, ZOO perturbs the input sequence $\mathbf{X}$ along individual coordinates, queries the model $f(\mathbf{X})$ to obtain predictions, and iteratively updates the perturbation δ using the estimated gradient. This query-based mechanism enables adversarial example generation without access to model internals.

C.4. Evaluation Metric Formulations

Traditional regression evaluation relies heavily on L_p -norm based metrics, such as Mean Squared Error (MSE) and Mean Absolute Error (MAE). While these metrics are theoretically sound for strictly convex optimization, they exhibit fundamental pathologies when used to assess adversarial robustness in Long-Term Time Series Forecasting (LTSF). To provide a rigorous evaluation of adversarial efficacy, we first delineate the theoretical deficiencies of point-wise metrics and then introduce trajectory-level alternatives.

C.4.1. THE PATHOLOGY OF POINT-WISE METRICS (L_p NORMS)

Standard forecasting metrics enforce a rigid, point-to-point comparison between the ground truth sequence $\mathbf{Y} = \{y_1, \dots, y_H\}$ and the predicted sequence $\hat{\mathbf{Y}} = \{\hat{y}_1, \dots, \hat{y}_H\}$. This "lock-step" formulation suffers from three critical limitations in the adversarial context:

1. The Lock-step Rigidity and Phase Sensitivity. Euclidean distance assumes a linear temporal alignment, comparing the t -th step of the prediction exclusively with the t -th step of the ground truth. As noted in time series mining literature (Ratanamahatana & Keogh, 2004), this assumption renders the metric brittle to temporal warping. Adversarial attacks operating in the frequency domain often induce phase shifts rather than additive noise. A perturbation that merely delays a peak by δ steps creates a misalignment that L_p norms interpret as a catastrophic amplitude error, failing to distinguish between meaningful structural corruption (shape destruction) and benign temporal misalignment.

2. The Double Penalty Effect. Originating from meteorological verification (Ebert & McBride, 2000), the "double penalty" paradox occurs when a predicted feature is structurally preserved but spatially or temporally displaced. MSE penalizes such displacement twice: once for missing the event at the correct time t (Miss Error), and again for predicting it at the incorrect time $t + \delta$ (False Alarm Error). Consequently, a trivial "safe" forecast that predicts a flat mean (smoothing) often yields a lower MSE than a structurally coherent but slightly phase-shifted prediction. In adversarial evaluation, this paradoxically incentivizes attacks that induce featureless smoothing over those that preserve trajectory structure but induce timing errors.

3. Directional Isotropy. L_p norms are isotropic, quantifying the magnitude of the error vector while ignoring its direction. In decision-critical applications such as energy dispatch or quantitative trading, the sign of the gradient (trend direction) is often more significant than absolute precision. An adversarial perturbation that reverses a trend (i.e., directional failure) constitutes a severe semantic error. However, MSE treats a trend reversal and a simple amplitude overshoot as equivalent if their Euclidean distances are identical, thereby failing to capture the logical validity of the forecasted trajectory.

Given these limitations, we adopt trajectory-level metrics to decouple shape deformation from temporal alignment and to explicitly measure directional integrity.

C.4.2. DYNAMIC TIME WARPING (DTW)

To address the lock-step limitation, we employ Dynamic Time Warping (DTW). For prediction sequences $\hat{\mathbf{Y}}, \hat{\mathbf{Y}}' \in \mathbb{R}^H$, DTW is defined as:

$$\text{DTW}(\hat{\mathbf{Y}}, \hat{\mathbf{Y}}') = \min_{\pi \in \mathcal{A}(H)} \sqrt{\sum_{(i,j) \in \pi} (\hat{Y}_i - \hat{Y}'_j)^2}$$

where $\mathcal{A}(H)$ denotes the set of admissible warping paths satisfying boundary conditions, monotonicity, and continuity constraints. DTW computes the Euclidean distance under optimal temporal alignment via dynamic programming with $\mathcal{O}(H^2)$ complexity.

By optimizing over all possible temporal alignments, DTW distinguishes genuine structural deformation from simple phase shifts. Consider phase shift attacks of the form $\hat{Y}'_i = \hat{Y}_{i-\tau}$, which merely delay predictions without altering trajectory characteristics. DTW captures such temporal misalignment. The quadratic cost accumulation provides sensitivity to amplitude-based attacks while maintaining interpretability as Euclidean distance under optimal alignment. The global optimization captures cumulative perturbation effects across the entire forecasting horizon, which is essential for LTSF evaluation.

For multivariate forecasting $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times F}$, we employ independent DTW:

$$\text{DTW-I}(\hat{\mathbf{Y}}, \hat{\mathbf{Y}}') = \frac{1}{F} \sum_{f=1}^F \text{DTW}(\hat{\mathbf{Y}}_{:,f}, \hat{\mathbf{Y}}'_{:,f})$$

C.4.3. SLOPE MISALIGNMENT ERROR (SME)

While DTW handles temporal shifts, it does not explicitly account for trend reversals if the warping path compensates for them. To capture directional dynamics, we employ Slope Misalignment Error (SME). Let $\Delta \hat{\mathbf{Y}} = [\hat{Y}_{i+1} - \hat{Y}_i]_{i=1}^{H-1}$ denote the first-order difference sequence. SME is defined as:

$$\text{SME}(\hat{\mathbf{Y}}, \hat{\mathbf{Y}}') = \sqrt{\frac{1 - \langle \Delta \hat{\mathbf{Y}}, \Delta \hat{\mathbf{Y}}' \rangle / (\|\Delta \hat{\mathbf{Y}}\|_2 \|\Delta \hat{\mathbf{Y}}'\|_2)}{2}}$$

where $\text{SME} \in [0, 1]$. Geometrically, this metric quantifies directional deviation in normalized gradient space, corresponding to chord distance on the unit hypersphere.

Operating in gradient space, SME directly quantifies trend reversals and rate-of-change perturbations. In numerous forecasting applications, directional accuracy assumes greater importance than absolute values. Trend reversal attacks of the form $\hat{\mathbf{Y}}' = -\hat{\mathbf{Y}}$ yield maximum SME values of 1.0, indicating complete directional opposition. The normalization ensures scale invariance: $\text{SME}(\alpha \hat{\mathbf{Y}}, \beta \hat{\mathbf{Y}}') = \text{SME}(\hat{\mathbf{Y}}, \hat{\mathbf{Y}}')$ for any positive scalars $\alpha, \beta > 0$, thereby isolating directional attacks from magnitude effects. First-order differencing functions as a high-pass filter, attenuating high-frequency noise while preserving trend characteristics.

C.4.4. METRIC COMPLEMENTARITY

DTW and SME measure orthogonal aspects of trajectory perturbation. DTW operates in ambient trajectory space $\mathbb{R}^H$, quantifying point-wise deviations under optimal temporal alignment to identify structural deformation. SME operates in normalized gradient space $\mathbb{S}^{H-2}$, quantifying angular deviations to identify logical trend failures. This fundamental distinction creates complementary sensitivity patterns. Constant offset attacks $\hat{\mathbf{Y}}' = \hat{\mathbf{Y}} + c$ induce uniform point-wise

deviations, yielding high DTW sensitivity, yet preserve gradients unchanged, resulting in SME insensitivity. Conversely, harmonic injection attacks simultaneously generate magnitude deviations and directional instability, producing high sensitivity in both metrics. By jointly reporting DTW and SME, we ensure that adversarial efficacy is assessed based on both geometric distortion and semantic trend misalignment.

Figure 8 provides synthetic illustrations to make the complementarity of DTW and SME concrete. Panel (a) shows a high-DTW, low-SME case where amplitude is scaled while trend direction is preserved, a pattern that point-wise metrics would penalize heavily yet fail to distinguish from directional failures. Panel (b) shows a low-DTW, high-SME case where slope sign is inverted while magnitude remains comparable, precisely the type of semantic corruption that DTW alone would underreport. Panel (c) confirms that these two distortion types occupy well-separated regions of the joint DTW-SME space, with phase shift occupying an intermediate position, supporting the use of both metrics for a comprehensive evaluation of adversarial efficacy.

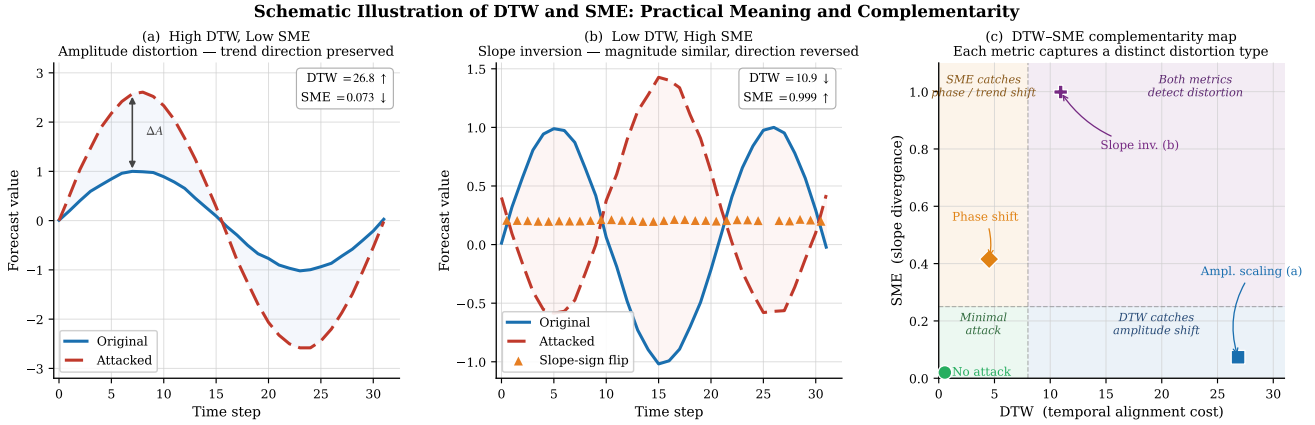


Figure 8. Synthetic Illustration of DTW and SME Complementarity. (a) Amplitude scaling yields high DTW and low SME, as trend direction is preserved. (b) Slope inversion yields low DTW and high SME, as magnitude remains similar while directional content is reversed. (c) DTW-SME space confirming that distinct distortion types occupy geometrically separated regions.

C.5. Cost Analysis of Sensitivity Profiling

We clarify the computational cost of the offline sensitivity profiling phase. While a naive frequency-domain traversal would imply iterating over all T time steps for both amplitude and phase dimensions, we leverage the **conjugate symmetry** property of the Discrete Fourier Transform (DFT) for real-valued time series. Specifically, for a real input signal, the Fourier coefficients satisfy $G[T - k] = \overline{G[k]}$. Consequently, perturbations need only be applied to the first $\lfloor T/2 \rfloor + 1$ frequency components to fully characterize the spectral sensitivity, as the negative frequency components are deterministically coupled.

Incorporating the calibration set size N_{cal} , the total number of queries Q_{sens} required for profiling a model with input length T and feature dimension F is calculated as:

$$Q_{\text{sens}} = N_{\text{cal}} \times F \times \underbrace{\left(\frac{T}{2} \times 2 \right)}_{\text{freqs} \times (\text{amp} + \text{phase})} \approx N_{\text{cal}} \times F \times T$$

where the factor of $T/2$ accounts for the positive frequency spectrum, and the factor of 2 accounts for independent amplitude and phase perturbations at each distinct frequency index.

Table 4 reports the query cost breakdown across benchmark datasets. In our experimental setup, we use a calibration budget of $N_{\text{cal}} = 50$ and sequence length $T = 96$. For the 7-channel ETT datasets ($F = 7$), the total offline profiling cost is:

$$Q_{\text{sens}} = 50 \times 7 \times 96 = 33,600 \text{ queries}$$

This aligns with the values reported in Table 4. Similarly, the 21-channel Weather dataset ($F = 21$) requires $Q_{\text{sens}} = 50 \times 21 \times 96 = 100,800$ queries.

The empirical results demonstrate that sensitivity profiling constitutes only 6.03% of the total experimental budget for ETTh1/ETTh2 datasets, 1.47% for ETTm1/ETTM2 datasets, and 4.83% for the Weather dataset. Crucially, this overhead is **amortized** across all attack instances—sensitivity maps are computed once per model and reused throughout the evaluation. For example, on ETTm1, the amortized cost per test instance is negligible ($33,600/2,848 \approx 11.8$ queries), validating that one-time sensitivity profiling represents a highly tractable preprocessing investment.

Table 4. Query cost breakdown for sensitivity profiling. Q_{sens} denotes the one-time offline profiling cost, Q_{attack} denotes the total online query budget across all test instances.

Dataset	F	Q_{sens}	Q_{attack}	Ratio (%)
ETTh1	7	33,600	557,000	6.03%
ETTh2	7	33,600	557,000	6.03%
ETTM1	7	33,600	2,285,000	1.47%
ETTM2	7	33,600	2,285,000	1.47%
Weather	21	100,800	2,088,800	4.83%

C.6. Cross-Dataset Transferability of Sensitivity Profiles

To empirically validate the domain-level calibration assumption stated in the threat model (Section 2.1), we conduct a cross-dataset transfer experiment within the ETT domain, using sensitivity profiles derived from ETTh1 validation data to attack ETTh2 test instances and vice versa. Table 5 reports DTW and SME under matched profiles (e.g., $h1 \rightarrow h1$) and cross-dataset profiles (e.g., $h2 \rightarrow h1$) across all seven architectures.

Table 5. Cross-dataset transferability of sensitivity profiles within the ETT domain. Notation $h_i \rightarrow h_j$ denotes profiles derived from ETTh i applied to attack ETTh j .

Model	h1→h1		h2→h1		h2→h2		h1→h2	
	DTW	SME	DTW	SME	DTW	SME	DTW	SME
Autoformer	32.485	0.166	32.368	0.164	25.226	0.291	24.880	0.282
DLinear	24.837	0.043	25.726	0.040	22.622	0.049	22.631	0.048
PatchTST	22.678	0.043	22.610	0.043	31.248	0.162	31.208	0.161
TimeMixer	24.479	0.038	23.755	0.035	28.292	0.078	27.657	0.082
TimeXer	36.016	0.035	36.747	0.035	45.642	0.098	43.406	0.095
TimesNet	63.106	0.108	61.336	0.115	66.807	0.290	66.998	0.289
iTransformer	35.580	0.046	36.791	0.046	43.201	0.111	43.485	0.113
Mean	34.169	0.068	34.190	0.068	37.577	0.154	37.181	0.153

Cross-dataset profiles yield attack performance comparable to matched profiles across all seven architectures, with negligible mean differences in both DTW (34.169 vs. 34.190 for $h1 \leftrightarrow h2$; 37.577 vs. 37.181 for $h2 \leftrightarrow h1$) and SME. These results suggest that sensitivity profiles derived from domain-relevant calibration data transfer effectively within the same application domain, supporting the practical viability of the domain-level calibration assumption.

C.7. Hyperparameter Analysis: SME Results

Figure 9 and Figure 10 present the SME counterparts to the DTW hyperparameter analyses in Section 4.3. Both figures confirm that the conclusions drawn from DTW generalize to SME: TSFAdv converges rapidly under varying query budgets and scales consistently under varying perturbation budgets, maintaining a clear advantage over all baselines across all four architectures.

D. Case Study

Figure 11 presents a representative attack instance on the ETTh dataset, illustrating that the adversarial perturbation remains imperceptible in the input history while inducing systematic directional deviations in the forecast horizon.

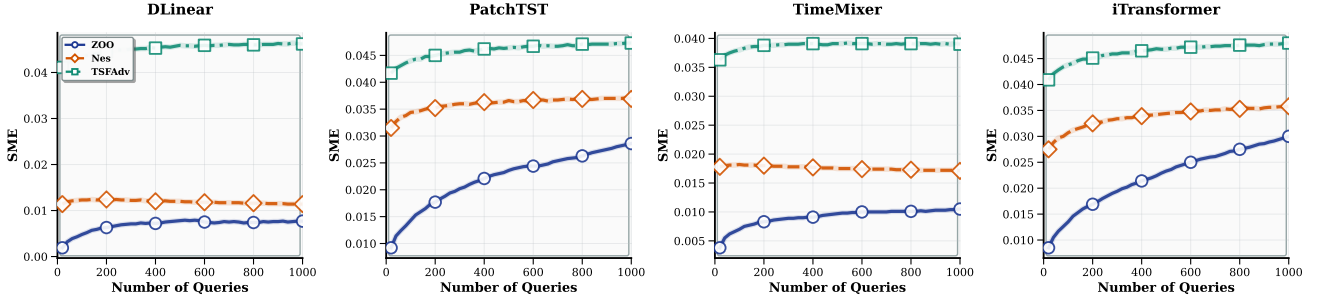


Figure 9. SME under Varying Query Budgets. SME versus query budget (20–1000 queries, $\epsilon = 0.1$), averaged across ETTh datasets. TSFAdv converges rapidly and maintains a consistent advantage over all baselines, mirroring the DTW convergence behavior in Figure 5.

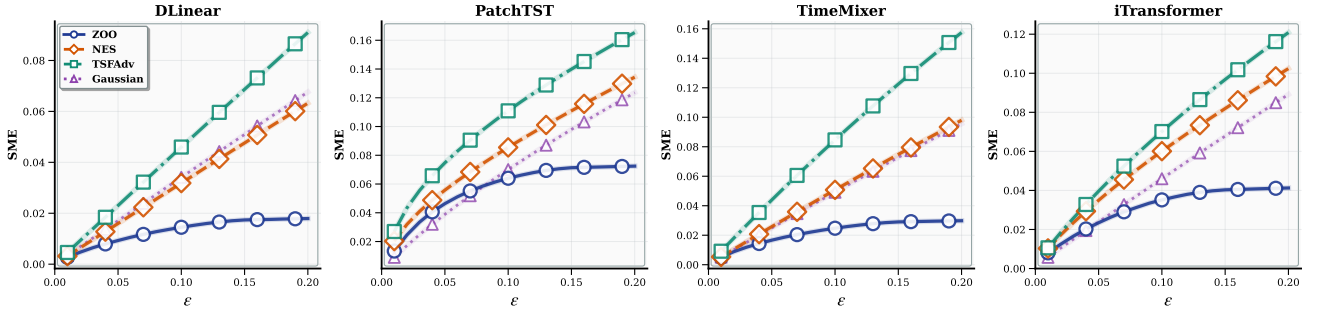


Figure 10. SME under Varying Perturbation Budgets. SME versus perturbation budget ($\epsilon \in [0.01, 0.2]$, fixed 200 queries), averaged across ETTh datasets. TSFAdv consistently achieves the highest SME with steeper scaling trends, mirroring the DTW results in Figure 4.

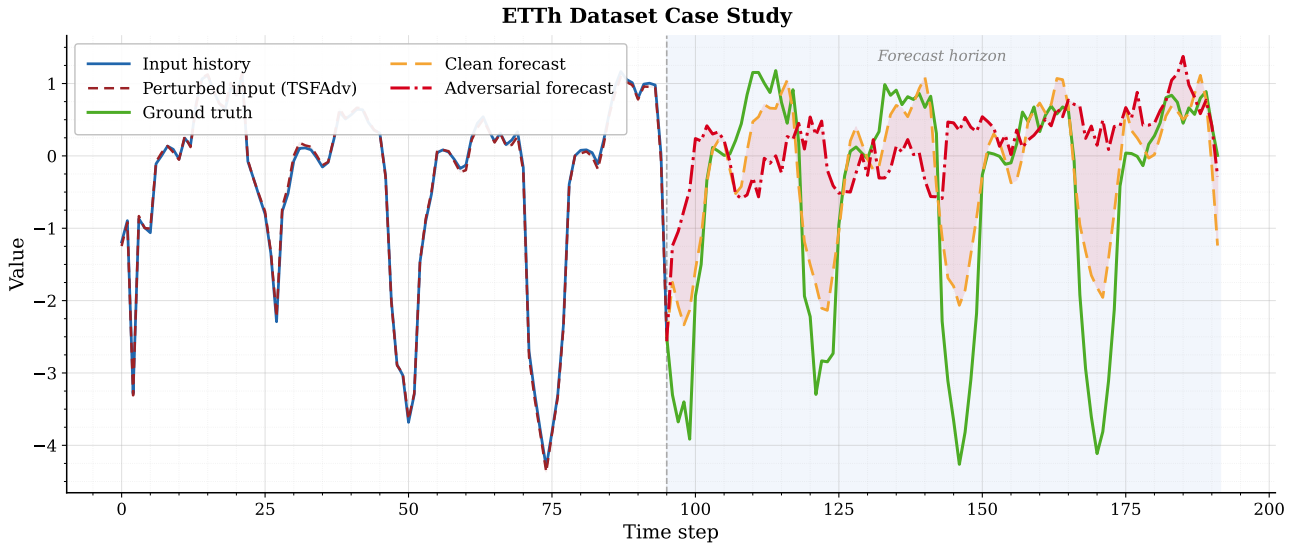


Figure 11. Case Study on ETTh Dataset. The perturbed input (red dashed) remains imperceptible within the input history (blue), while the adversarial forecast (red dash-dot) exhibits pronounced directional deviations from the clean forecast (orange dashed) and ground truth (green) within the forecast horizon (shaded blue region).

E. Detailed Implementation of Defense Mechanisms

To comprehensively evaluate the robustness of Long-Term Time Series Forecasting (LTSF) models against the proposed TSFAdv, we implement three representative defense strategies adapted from the computer vision domain: Randomized Smoothing, Magnitude-based Pruning, and Adversarial Training. These methods serve as countermeasures to mitigate the impact of adversarial perturbations during the inference or training phases.

E.1. Randomized Smoothing

We adopt the certified robustness framework proposed by Cohen et al. (Cohen et al., 2019), which constructs a smoothed function g from a base regressor f . The core premise is that convolving the base model with isotropic Gaussian noise suppresses high-frequency adversarial perturbations, thereby enhancing stability.

Formally, for an input time series $X \in \mathbb{R}^{T \times F}$, the smoothed prediction $\hat{Y}_{smooth}$ is estimated via Monte Carlo sampling. We generate $N = 100$ noise-corrupted samples for each input instance:

$$\hat{Y}_{smooth}(X) = \frac{1}{N} \sum_{i=1}^N f(X + \delta_i), \quad \delta_i \sim \mathcal{N}(0, \sigma^2 \mathbf{I}),$$

where δ_i represents the stochastic noise vector sampled from a Gaussian distribution with zero mean and variance σ^2 . We evaluate the defense across three noise magnitude levels, $\sigma \in \{0.05, 0.1, 0.15\}$, to analyze the trade-off between robustness and clean accuracy. This method transforms the deterministic forecasting model into a stochastic ensemble during inference, aiming to "drown out" the specific spectral patterns exploited by the attacker.

E.2. Magnitude-based Pruning

We employ unstructured Magnitude-based Pruning (Han et al., 2015) to investigate whether reducing model over-parameterization alleviates susceptibility to adversarial attacks. This hypothesis posits that sparse models may exhibit lower sensitivity to minute input perturbations by eliminating redundant, non-robust feature pathways.

The pruning process is executed in a two-stage pipeline:

1. **Sparsification:** We compute a global threshold τ_p determined by the 20th percentile of all absolute weight values $|\mathbf{W}|$ in the network. A binary mask $\mathbf{M}$ is applied to the weights, such that:

$$\mathbf{W}' = \mathbf{W} \odot \mathbf{M}, \quad \text{where } M_{ij} = \mathbb{I}(|W_{ij}| \geq \tau_p).$$

This operation permanently removes 20% of the model parameters with the lowest magnitudes.

2. **Fine-tuning:** To recover the forecasting performance degraded by pruning, the sparsified model is fine-tuned for 3 epochs on the original training set with the mask $\mathbf{M}$ frozen, ensuring the structural sparsity is maintained.

E.3. Adversarial Training (Gaussian Augmentation)

To induce robustness against input variations during the optimization phase, we implement an Adversarial Training regime (Madry et al., 2018). While standard adversarial training often employs gradient-based attacks (e.g., PGD), we utilize Gaussian noise augmentation as a stochastic approximation of adversarial perturbations. This approach compels the model to learn representations that are invariant to high-frequency jitter, which is conceptually aligned with defending against spectral attacks.

We formulate the training objective as a min-max optimization problem where the inner maximization is approximated by random noise injection. For a clean training batch $\{(X_{clean}, Y)\}$, we generate augmented adversarial examples X_{adv} via:

$$X_{adv} = X_{clean} + \xi, \quad \xi \sim \mathcal{N}(0, \epsilon^2 \mathbf{I}),$$

where the perturbation magnitude is constrained by $\epsilon = 0.1$. The model is trained on a mixed dataset comprising a **1:1 ratio** of clean samples X_{clean} and perturbed samples X_{adv} . This data augmentation strategy effectively regularizes the decision boundary (or regression manifold), encouraging the model to smooth its output landscape in the vicinity of the training data distribution.